

AI-BFSI Internship Project Documentation

Technology Stack – Subscription Waste Detector

1. Introduction

This document outlines the **technology stack** used in the Subscription Waste Detector project. It provides a structured overview of the tools, frameworks, and libraries that enable the system's functionality. The stack is carefully selected to support expense data processing, subscription detection, dashboard visualization, and AI-powered recommendations.

2. Project Type

- **AI-powered subscription expense analysis web application**
- Designed to help users identify recurring subscriptions, visualize spending, and receive cost-saving recommendations.

3. Frontend Technologies

- **Streamlit**
 - Provides the web application framework.
 - Enables interactive dashboards and user-friendly interfaces.
- **Plotly**
 - Used for dynamic charting and visualization.
 - Supports interactive graphs for subscription spending analysis.
- **HTML/CSS (via Streamlit components)**
 - Enhances UI customization.
 - Provides styling for dashboard elements.

4. Backend Technologies

- **Python**
 - Core programming language for application logic.
 - Handles data ingestion, processing, and integration with external APIs.

5. Libraries

- **Pandas**
 - Data manipulation and analysis.
 - Detects recurring transactions and categorizes subscriptions.
- **Plotly Express**
 - Simplified interface for creating charts.
 - Used for subscription summaries and monthly spending visualization.
- **OpenPyXL**

- Supports Excel file uploads.
 - Enables reading and parsing of spreadsheet data.
- **Groq SDK**
 - Facilitates communication with the Groq API.
 - Sends subscription data for AI-powered recommendations.

6. AI Integration

- **Groq API**
 - Provides AI-powered recommendations.
 - Processes subscription data and returns cost-saving suggestions.
- **Llama Model (via Groq)**
 - Powers the recommendation engine.
 - No custom training performed — strictly API-based inference.

7. Data Storage

- **Current Implementation:**
 - Uploaded files (CSV/Excel)
 - In-memory processing
 - Session-based operation within Streamlit
- **Not Implemented:**
 - Permanent storage
 - Database integration (SQLite, Firebase)
 - Cloud synchronization

8. Architecture Overview

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text
User
↓
Streamlit UI
↓
Python Processing
↓
Pandas Analysis
↓
Subscription Detection
↓
Groq API (AI Recommendations)
↓
Dashboard Visualization

```

9. Future Enhancements (Not Implemented Yet)

The following technologies may be introduced in future versions:

- Database integration (SQLite, Firebase)
- User authentication systems
- Multi-user support

- Bank API integration for automatic statement fetching
- Predictive analytics and spending forecasting
- Multi-currency support
- Cloud synchronization

10. Conclusion

The Subscription Waste Detector leverages a **lightweight, efficient technology stack** centered on Python, Streamlit, Pandas, Plotly, and Groq API. This stack ensures seamless expense analysis, subscription detection, and AI-powered recommendations, while remaining